Sir Syed University of Engineering & Technology
Faculty of Computing and Applied Sciences
Department of Computer Science & Information Technology
Mid Term Examinations (Fall 2023) SAMPLE SOLUTION

Course Code with Title	**CS-215T: Computer Organization & Assembly Language**	**Program**	Bachelor of Science in Computer Science BS(CS)
Instructor(s)	**Mr. Shakir Karim / Dr. Abdul Hameed Pitafi**	**Semester**	3^{rd}
Start date & Time	Dec 08, 2023 at 9:15 AM	**End Time**	10:45 AM
Maximum Marks	30		

<u>IMPORTANT INSTRUCTIONS</u>:

Read the following Instructions carefully:

- Attempt All Questions.
- Use of calculator (simple/scientific/programmable) is not allowed.
- Return the Question Paper along with the Answer Script.

Q.No.1	(CLO_1): (Cognitive Level C2, i.e., Understanding)	(PLO_1: Academic Education) **(05 + 05 Marks)**

Consider the following set of the Intel x86 Assembly Language instructions:

MOV [110A], BX
PUSH AX
SHR AL, CL
ADD AL, CL
AAA

Identify the following, for the each given instruction:

(a) The addressing modes.

(b) The memory addresses with the contents of memory location, after execution.

Assume that the values in the registers as:
DS=AC80h, **SS=50F0h**, **SP=FFFDh, AX=1734h, BX=1254h, DI=560Dh,** and **CX=0005**.

Answer

Set of Instructions	(a)	(b)	
	Addressing Modes (1)	**Memory Address (0.5)**	**Memory Contents (0.5)**
MOV [110A], BX	Direct	DS:110A=AC800+110A AD90Ah AD90Bh	 54h 12h
PUSH AX	Stack / Register	SS:SP=50F00+(FFFD-2) =50F00+FFFB=60EFBh 60EFBh 60EFCh	 34h 17h

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Set of Instructions	(b) Addressing Modes (1)	(b) Memory Address (0.5)	Memory Contents (0.5)
SHR AL, CL	Register	AL register	AL = 01h
ADD AL, CL	Register	AL register	AL = 06h
AAA	ASCII Adjustment after Addition, Used in combination with ADD instruction (can be left blank)	AL register (can be left blank)	AL = 06h

Q.No.2	(CLO_1): (Cognitive Level C2, i.e., Understanding) (PLO_1: Academic Education) **(05 Marks)**

Express, step-by-step, the standard process of signed multiplication to multiply **(-8)** and **(+7)**. Assume 4-bits registers to hold signed numbers.

Answer

-8 = 1000 **+7 = 0111**

Iteration #	$A_3A_2A_1A_0$	$Q_3Q_2Q_1Q_0$	Q_{-1}	$M_3M_2M_1M_0$	Remarks	
Initializing	0 0 0 0	1 0 0 0	0	0 1 1 1		1
Iteration #1	0 0 0 0	0 1 0 0	0	0 1 1 1	SAR	1
Iteration #2	0 0 0 0	0 0 1 0	0	0 1 1 1	SAR	1
Iteration #3	0 0 0 0	0 0 0 1	0	0 1 1 1	SAR	1
Iteration #4	1 0 0 1 **1 1 0 0**	0 0 0 1 **1 0 0 0**	0 1	0 1 1 1	A=A-M SAR	1

-8 x +7 = -56 = 11001000_2 (2's complement of 56)

Q.No.3	(CLO_1): (Cognitive Level C2, i.e., Understanding) (PLO_1: Academic Education) **(05 Marks)**

Represent 112.0625_{10} in **IEEE 754 double precision** format by showing all the necessary steps involved in the process.

Answer:
Since double precision is 64 bit format.
(**01** Sign bit, **11** Biased Exponent bits, **52** Significand bits)
Here, Bias = $2^{11-1}-1 = 2^{10}-1 = 1024-1 = 1023$
$112.0625_{10} = 1110000.0001_2 = 1.1100000001 \times 2^6$
Sign bit = **0** since number is +ve
Biased Exponent = True Exponent + Bias = 6 + 1023 = 1029_{10} = $\mathbf{10000000101}_2$
Significand = $\mathbf{1100000001000000000000000000000000000000000000000000}_2$
112.0625_{10} = $\mathbf{0\ 10000000101\ 1100000001000000000000000000000000000000000000000000}_2$
= **405C040000000000h**

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Q.No.4	(CLO_1): (Cognitive Level C2, i.e., Understanding) (PLO_1: Academic Education) **(06 Marks)**

Rewrite the following set of loop instructions into a separate procedure using the Intel x86 Assembly Language. The name of the procedure should be myProc.

```
void myProc (void)
{       int AL = 0;   int CX;
             for ( CX = 10;   CX > 0;   CX - - )
             AL = AL + 1;                          }
```

Answer

myProc PROC	**1**
MOV AL, 0	**1**
MOV CX, 10	**1**
AGAIN:	
ADD AL,1	$\left.\right\}$ **1**
LOOP AGAIN	
RET	**1**
myProc ENDP	**1**

Q.No.5	(CLO_1): (Cognitive Level C2, i.e., Understanding) (PLO_1: Academic Education) **(01 + 01 + 01 + 01 Marks)**

Consider a processor which has 256 instructions and a total of 16 registers. **Express** the given instructions in the respective machine code by referring to Table 1 and the instruction set summary (for *ADD* and *SUB* instructions).

(a) SUB [BP+SI], AL **(b) ADD [205A], DL**

(c) ADD DS, 5434H **(d) SUB [DI], BL**

Table 1: R/M and MOD values

MOD / R/M	00 0-bit displacement	01 8-bit displacement	10 16-bit displacement	11 w = 0	11 w = 1
000	[BX+SI]	[BX+SI]+D8	[BX+SI]+D16	AL	AX
001	[BX+DI]	[BX+DI]+D8	[BX+DI]+D16	CL	CX
010	[BP+SI]	[BP+SI]+D8	[BP+SI]+D16	DL	DX
011	[BP+DI]	[BP+DI]+D8	[BP+DI]+D16	BL	BX
100	[SI]	[SI]+D8	[SI]+D16	AH	SP
101	[DI]	[DI]+D8	[DI]+D16	CH	BP
110	DIRECT	[BP]+D8	[BP]+D16	DH	SI
111	[BX]	[BX]+D8	[BX]+D16	BH	DI

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8086/8088 Instruction Set Summary

Mnemonic and Description	Instruction Code			
ARITHMETIC	7 6 5 4 3 2 1 0	7 6 5 4 3 2 1 0	7 6 5 4 3 2 1 0	7 6 5 4 3 2 1 0
ADD = Add:				
Reg./Memory with Register to Either	0 0 0 0 0 0 d w	mod reg r/m		
Immediate to Register/Memory	1 0 0 0 0 0 s w	mod 0 0 0 r/m	data	data if s: w = 01
Immediate to Accumulator	0 0 0 0 0 1 0 w	data	data if w = 1	
SUB = Subtract:				
Reg./Memory and Register to Either	0 0 1 0 1 0 d w	mod reg r/m		
Immediate from Register/Memory	1 0 0 0 0 0 s w	mod 1 0 1 r/m	data	data if s w = 01
Immediate from Accumulator	0 0 1 0 1 1 0 w	data	data if w = 1	

Answer

(a) SUB [BP+SI], AL

Register to Register/Memory Addressing
001010dw mod reg r/m
Reg is AL, R/M is [BP+SI], mod is 00 (memory without displacement)
d = 0 since Reg is source, w = 0 since Reg is 8 bit register
001010 0 0 00 000 010 = 2802h

(b) ADD [205A], DL

Register to Register/Memory Addressing
000000dw mod reg r/m
Reg is DL, R/M is [205A], mod is 00 (memory without displacement) DIRECT
d = 0 since Reg is source, w = 0 since Reg is 8 bit register
000000 0 0 00 010 110 01011010 00100000 = 0016 5A20h

(c) ADD DS, 5434H

No such instruction available in instruction set – ILLEGAL INSTRUCTION

(d) SUB [DI], BL

Register to Register/Memory Addressing
001010dw mod reg r/m
Reg is BL, R/M is [DI], mod is 00 (memory without displacement)
d = 0 since Reg is source, w = 0 since Reg is 8 bit register
001010 0 0 00 011 101 = 281Dh